Problem 3: Life (10 × 2 = 20 points) [Codeforces]

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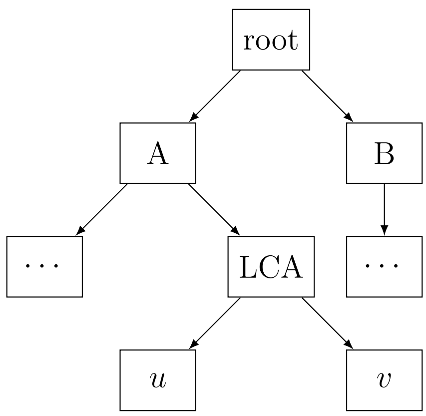
The wintry weather has set in and taken its toll. You do not have a winter jacket, so you only go outside from 2-5pm. As such, you are unable to attend any parties, breakfasts, late-night club sessions, or half of your classes, and you have realized your life is going rather poorly. Others may think this problem has an easy solution (buy a winter jacket!) but you know that this is not the case, and scheme so that your life may be better.

Part 1 (10 points)

You decide to represent life as a decision tree. You are now at the root state. At each state, you face several choices; depending on your decision, you move to one of the possible child states. With this tree in hand, you could compute the optimal path through life.

Unfortunately, you do not have the tree itself. Stricken by decision paralysis, you refuse to approximate it, and instead opt to construct it precisely using your knowledge of the lowest common ancestor (LCA) of every pair of states: the deepest state in the tree that serves as an ancestor to both. In other words, for each pair of states (u, v), you know the last shared state: the point just before the crucial decision that led to either u or v. Consult the following diagram.



Given these pairwise lowest common ancestors, reconstruct the tree.

Input

The first line contains an integer N. The following N lines contain an $N \times N$ grid l where $l_{i,j}$ denotes the lowest common ancestor of states i and j. That is, $l_{i,j}$ is the j'th integer on the $(i+1)$'th line of the input.

Output

Print N space-separated integers, where the i'th integer is the parent state of state i, or -1 if state i is the root.

Constraints

You can assume that $N \leq 1000$.

Partial Credit

You will get 5 points for solving cases where $N \leq 100$.

Example

Input:

```
5
1 5 5 4 5
5 2 3 5 5
5 3 3 5 5
4 5 5 4 5
5 5 5 5 5
```

Output:

```
4 3 5 5 -1
```

Explanation: This input is the same as the diagram on the previous page.

Part 2 (10 points)

It turns out that your previous matrix was wrong! What a disaster! You consider recalculating a correct one, but then you remember that your decision tree has 1000 states, meaning there are 499,500 lowest common ancestors to determine. (No wonder you made mistakes last time.)

Fortunately, you realize that you don't actually need to know every pairwise lowest common ancestor to reconstruct the tree. Instead, you can follow a more efficient procedure:

- Choose a pair of states (u, v) whose lowest common ancestor you deem important to know.
- Determine the lowest common ancestor of u and v.
- Repeat this process until you have recovered the entire original tree.

Clearly, the time you take is proportional to the number of pairwise lowest common ancestors you have to determine, so you want to minimize that. Helping you are two facts:

- There is a single ground truth tree, so the structure of the tree does not change based on your actions.
- The tree in question is a decision tree, so each state has very few children. In fact, the root has a maximum of 3 children, and every other state has at most 2 children.

Reconstruct the tree using this procedure. There are $N \leq 1000$ states, and you get $Q = 25000$ queries to reconstruct the tree. Each query is a determination of the lowest common ancestor of some pair of states (u, v).

Interaction

This is an interactive problem. Your program will interact with a grader program.

First, read a single line of input which contains N, the number of nodes, and Q, the maximum number of queries you are allowed to ask.

After this, you should make some queries to the grader. To query the lowest common ancestor of some pair of nodes (X, Y), print `? X Y` and flush the output. Then, read in the lowest common ancestor from standard input. If this answer is -1, please terminate your program immediately.

When you have determined the structure of the tree, print `! p1 p2 [...] pn`, where p_i is the parent of node i, or -1 if i is the root. For example, in the example, you would output `! 4 3 5 5 -1`. Then end your program.

Sample Interaction

Below is an example of an interaction in a testcase where $N = 3$, node 1 is the root, and nodes 2 and 3 are its children.

```
You             Grader
                3 25000     //  N, Q
? 1 2
                1           // LCA(1, 2) = 1
? 2 3
                1           // LCA(2, 3) = 1
! -1 1 1                    // You provide the correct answer.
```